UNIVERSITY PHYSICS REFERENCE SERIES • VOLUME 4

Rotational Dynamics & Angular Momentum

Rigid Body Kinematics, Torque & Newton's Second Law for Rotation, Moments of Inertia & The Parallel Axis Theorem, Rotational Kinetic Energy, Rolling Without Slipping, Vector Angular Momentum Conservation, and Gyroscopic Precession Mechanics

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Synthesized From:

• Halliday, Resnick, Walker — *Fundamentals of Physics* (Chapters 10 & 11: Rotation & Rolling)

• R. Shankar (Yale University) — *Fundamentals of Physics I: Mechanics, Relativity, and Thermodynamics* (Chapters 9, 10, 11)

• University Lecture Slides — *Chapter 4: Rotation of a Rigid Body*

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Includes: Scalable SVG Vector Visualizations • 5 Calculation Fast-Tracks • 4 Complete Worked Problems

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Part I: Pure Rotation Kinematics & Mass Distribution

Chapter 1: Angular Kinematics & Vector Rotation

In translational kinematics, a point particle's motion is governed by position **r**(t), linear velocity **v**(t) = d**r**/dt, and linear acceleration **a**(t) = d**v**/dt. For an extended **rigid body**—an object whose constituent particles remain at strictly invariant mutual distances—every point on the body sweeps through identical angles in equal time intervals when rotating about a fixed axis. Hence, the entire body's rotational state is completely characterized by single angular state variables.

1.1 Angular Coordinates, Radian Definition & Angular Velocity

Consider a rigid body constrained to rotate about a fixed axis perpendicular to the page (the z-axis). Let a reference line fixed within the body make an angle θ relative to a fixed spatial reference axis (e.g. the positive x-axis). The angular displacement Δθ for a point located at perpendicular distance r from the axis when it sweeps an arc length s is:

$$\theta = \frac{s}{r} \text{ [radians]}, \quad \text{where} \quad 1 \text{ rev} = 360^\circ = 2\pi \text{ rad} \approx 6.28318 \text{ rad}$$

The instantaneous **angular velocity** ω and instantaneous **angular acceleration** α are defined as the first and second time derivatives:

$$\omega = \frac{d\theta}{dt} \text{ [rad/s]}, \qquad \alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} \text{ [rad/s}^2\text{]}$$

For periodic rotation at constant angular speed ω, the motion repeats every period T (seconds per revolution), corresponding to a cyclic frequency f (revolutions per second, Hz):

$$f = \frac{1}{T}, \qquad \omega = 2\pi f = \frac{2\pi}{T}$$

1.2 Vector Nature of Rotational Variables & The Right-Hand Rule

Angular velocity **ω** and angular acceleration **α** are true **pseudovectors** (axial vectors) pointing along the axis of rotation:

> **▶ Right-Hand Rule for Angular Quantities**
>
> Curl the four fingers of your right hand in the direction of rotational motion along the circumference. Your outstretched right thumb points unambiguously in the direction of the angular velocity vector **ω**. If ω is increasing, **α** points in the same direction as **ω**; if slowing down (braking), **α** points in the exact opposite direction.

1.3 Tangential vs. Centripetal (Radial) Linear Accelerations

While all particles in the rigid body share identical ω and α, their linear trajectories depend explicitly on their radial distance r from the axis:

Tangential Speed: $$v_t = \omega r$$

Tangential Acceleration (rate of speed change): $$a_t = \frac{dv_t}{dt} = \alpha r$$

Radial / Centripetal Acceleration (rate of direction change): $$a_c = \frac{v_t^2}{r} = \omega^2 r$$

Because the tangential and radial vectors are strictly mutually perpendicular ($a_t \perp a_c$), the total linear acceleration magnitude of any particle on the body is:

$$a_{total} = \sqrt{(a_t^2 + a_c^2)} = \sqrt{[(\alpha r)^2 + (\omega^2 r)^2]} = r\sqrt{(\alpha^2 + \omega^4)}$$

1.4 Rotational Kinematics Under Constant Angular Acceleration (α = const)

Integrating α = const yields four kinematic equations completely isomorphic to linear 1D constant-acceleration motion:

Translational Variable / Formula	Rotational Counterpart	Rotational Equation (α = const)	Missing Variable
Position x → Velocity v	Angle θ → Velocity ω	$\omega = \omega_0 + \alpha t$	$\theta - \theta_0$
$x - x_0 = \frac{1}{2}(v_0 + v)t$	$\theta - \theta_0 = \frac{1}{2}(\omega_0 + \omega)t$	$\theta = \theta_0 + \frac{1}{2}(\omega_0 + \omega)t$	α
$x = x_0 + v_0 t + \frac{1}{2}at^2$	$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$	$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$	ω
$v^2 = v_0^2 + 2a(x - x_0)$	$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$	$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$	t

Chapter 2: Moment of Inertia & The Parallel Axis Theorem

In linear dynamics, **mass** M is the intrinsic measure of translational inertia—a body's reluctance to change its linear velocity. In rotational dynamics, mass alone does not suffice: an identical mass distributed far from the rotation axis is vastly more difficult to spin up than when concentrated near the axis. This geometric mass distribution is captured by the **Moment of Inertia** (rotational inertia) I.

2.1 Discrete vs. Continuous Rotational Inertia

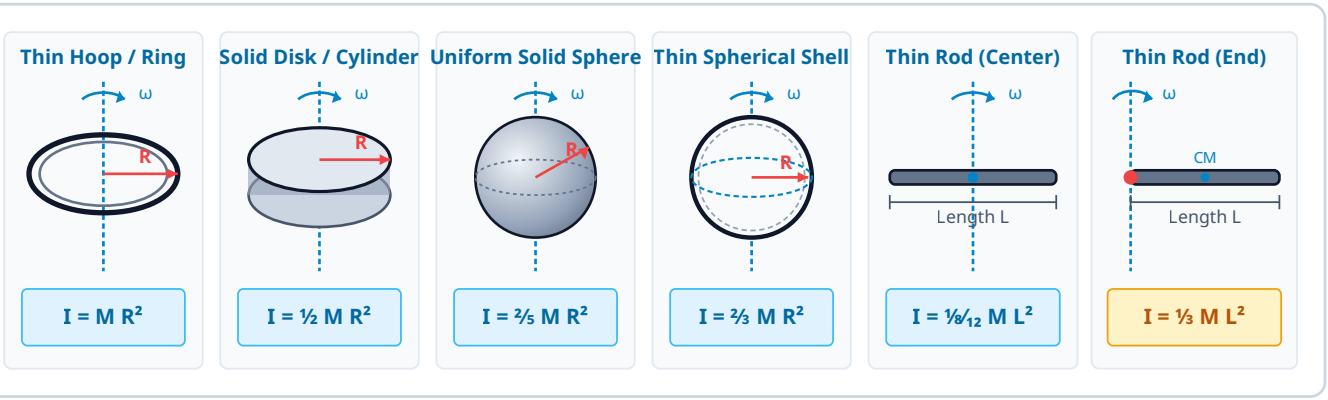
For a system of N discrete point particles with masses m_i at perpendicular distances r_i from the axis of rotation:

$$I = \sum_{i=1}^{N} m_i r_i^2 \ [kg \cdot m^2]$$

For a continuous solid body of total mass M with mass density ρ(**r**), the summation transitions into a volume integral:

$$I = \int r_{\perp}^{2}\, dm = \int r_{\perp}^{2}\, \rho\, dV$$

Figure 1: Geometric Atlas of Rotational Inertia (Standard Geometries & Principal Axes)



2.2 Standard Moments of Inertia Formula Table

Body Geometry	Rotation Axis Location	Moment of Inertia I	Dimensionless Factor c = I / (MR²)
Thin Cylindrical Hoop / Ring	Central symmetry axis	**$M R^2$**	c = 1.000
Uniform Solid Disk / Cylinder	Central cylinder axis	**$\frac{1}{2} M R^2$**	c = 0.500
Thick Hollow Cylinder	Central cylinder axis (radii R_1, R_2)	**$\frac{1}{2} M (R_1^2 + R_2^2)$**	$c = \frac{1}{2}(1 + R_1^2/R_2^2)$
Uniform Solid Sphere	Any central diameter axis	**$\frac{2}{5} M R^2$**	c = 0.400
Thin Hollow Spherical Shell	Any central diameter axis	**$\frac{2}{3} M R^2$**	c = 0.667
Uniform Thin Rod (Length L)	Perpendicular axis through Center of Mass	**$\frac{1}{12} M L^2$**	$c = \frac{1}{12}$ (relative to L^2)
Uniform Thin Rod (Length L)	Perpendicular axis through End Tip	**$\frac{1}{3} M L^2$**	$c = \frac{1}{3}$ (relative to L^2)
Flat Rectangular Plate (a × b)	Perpendicular axis through center	**$\frac{1}{12} M (a^2 + b^2)$**	$c = \frac{1}{12}$ (relative to diagonal)

2.3 The Parallel Axis Theorem (Steiner's Theorem)

If the moment of inertia I_{cm} about an axis passing through the **Center of Mass (CM)** is known, the moment of inertia I about any other parallel axis displaced by a perpendicular distance d is given by:

$$I = I_{cm} + M d^2$$

▶ Rigorous Proof of the Parallel Axis Theorem

Choose coordinate origin at the Center of Mass, such that $\int x\,dm = 0$ and $\int y\,dm = 0$. Let the new parallel axis be displaced along the xy-plane to coordinates (x_0, y_0) where $d^2 = x_0^2 + y_0^2$. The squared distance of mass element dm from the new axis is: $r'^2 = (x - x_0)^2 + (y - y_0)^2 = (x^2 + y^2) - 2x_0\,x - 2y_0\,y + (x_0^2 + y_0^2)$. Integrating over the entire body:

$$I = \int r'^2\,dm = \int (x^2 + y^2)\,dm - 2x_0 \int x\,dm - 2y_0 \int y\,dm + (x_0^2 + y_0^2) \int dm$$

By definition of CM, the cross-terms $\int x\,dm$ and $\int y\,dm$ vanish identically! The first term is I_{cm}, and $\int dm = M$. Therefore, $\mathbf{I = I_{cm} + M\,d^2}$. Q.E.D.

2.4 Perpendicular Axis Theorem & Radius of Gyration

Perpendicular Axis Theorem: For any flat 2D planar laminar sheet lying entirely in the xy-plane:

$$\mathbf{I_z = I_x + I_y}, \quad \text{since} \quad \int (x^2 + y^2)\,dm = \int y^2\,dm + \int x^2\,dm$$

Radius of Gyration (k): The effective distance from the axis where the entire mass M could be concentrated as a thin hoop without altering I:

$$\mathbf{I = M\,k^2} \quad \Rightarrow \quad \mathbf{k = \sqrt{(I / M)}}$$

⚡ FAST-TRACK 1 **Parallel Axis Mental Shift & Common Pitfalls** 2-Second Rule

Critical Pitfall: You CANNOT apply the parallel axis theorem directly between two non-CM points! For example, shifting from one end of a rod to another point via $I_2 = I_1 + M\,d^2$ yields completely erroneous answers!

Mental Shortcut: Always pivot through the Center of Mass in two steps:

$$\mathbf{I_{new} = (I_{old} - M\,d_{old}^2) + M\,d_{new}^2 = I_{cm} + M\,d_{new}^2}$$

Example: Rod End Tip: d = L/2 from CM ⇒ $I_{end} = \frac{1}{12}\,M\,L^2 + M\,(L/2)^2 = \frac{1}{12}\,M\,L^2 + \frac{1}{4}\,M\,L^2 = \mathbf{\frac{1}{3}\,M\,L^2}$ in 2 seconds of mental arithmetic!

Part II: Rotational Dynamics, Torque & Work-Energy

Chapter 3: Torque & Newton's Second Law for Rotation

Force produces linear acceleration (**F** = m**a**). The rotational counterpart of force is **Torque** (τ, from Latin *torquere*, to twist), which measures the quantitative capacity of an applied force to induce angular acceleration about a chosen origin or pivot.

3.1 Vector Definition of Torque

For a force **F** applied at position vector **r** measured relative to a chosen reference point O:

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} \ [\mathrm{N \cdot m}]$$

The scalar magnitude of torque can be evaluated via three completely equivalent geometric representations:

$$\tau = r F \sin\varphi = r (F \sin\varphi) = r F_t = (r \sin\varphi) F = r_\perp F$$

where φ is the angle between **r** and **F**, F_t is the tangential force component, and $\mathbf{r}_\perp = \mathbf{r} \sin \boldsymbol{\varphi}$ is the **moment arm** (lever arm)—the perpendicular distance from the rotation pivot to the line of action of the force.

3.2 Newton's Second Law for Rotation: ∑τ = I α

For a rigid body rotating about a fixed axis through its center of mass (or any fixed axis):

$$\sum \tau_{\mathrm{ext}} = I \alpha$$

Every internal particle experiences mutual action-reaction pairs ($\mathbf{F}_{ij} = -\mathbf{F}_{ji}$). Because these internal forces act along the line joining the particles, their net internal torque cancels identically: $\sum \tau_{\mathrm{internal}} = 0$. Hence, only external torques accelerate the body!

⚡ FAST-TRACK 2 **Massive Pulley Equivalent Linear Inertia Shortcut** 5 Seconds

Standard Method (Slow): Setting up three simultaneous coupled equations ($T_1 - m_1 g = m_1 a$, $m_2 g - T_2 = m_2 a$, $(T_2 - T_1) R = I\alpha$), solving 3 × 3 matrix systems.

⚡ **Fast Hand Calculation Shortcut:** Treat the rotating pulley as an **effective linear mass**:

$$m_{eff} = \frac{I}{R^2} \quad \text{(For a solid disk pulley: } m_{eff} = \tfrac{1}{2} M_p\text{)}$$

The linear acceleration of any Atwood machine or coupled pulley system is immediately:

$$a = \frac{\text{Net Driving Force}}{\text{Total Inertia}} = \frac{(m_2 - m_1) g}{m_1 + m_2 + \frac{I}{R^2}} = \frac{(m_2 - m_1) g}{m_1 + m_2 + \frac{1}{2} M_p}$$

Solves complex coupled Atwood systems in a single line of calculation!

Chapter 4: Rotational Work, Kinetic Energy & Power

The kinetic energy of an extended body is the sum of the kinetic energies of all its constituent particles: $K = \sum \frac{1}{2} m_i v_i^2 = \sum \frac{1}{2} m_i (\omega r_i)^2 = \frac{1}{2} (\sum m_i r_i^2) \omega^2$. This establishes the fundamental formula for **Rotational Kinetic Energy**:

$$K_{rot} = \tfrac{1}{2} I \omega^2 \text{ [Joules]}$$

4.1 Work Done by a Torque & The Rotational Work-Energy Theorem

When a torque τ rotates a rigid body through an infinitesimal angular displacement $d\theta$, the incremental work performed is:

$$dW = \tau \cdot d\theta = \tau \, d\theta \quad \Rightarrow \quad W = \int_{\theta_i}^{\theta_f} \tau \, d\theta$$

Using Newton's second law $\tau = I \alpha = I \, (d\omega/dt) = I \, \omega \, (d\omega/d\theta)$:

$$W_{net} = \int_{\theta_i}^{\theta_f} \left(I \, \omega \, \frac{d\omega}{d\theta}\right) d\theta = \int_{\omega_i}^{\omega_f} I \, \omega \, d\omega = \tfrac{1}{2} I \omega_f^2 - \tfrac{1}{2} I \omega_i^2 = \Delta K_{rot}$$

Instantaneous Rotational Power: The time rate of doing rotational work is:

$$P = \frac{dW}{dt} = \tau \frac{d\theta}{dt} = \tau \cdot \omega \text{ [Watts]}$$

Part III: Combined Motion & Conservation Principles

Chapter 5: Rolling Without Slipping: Dynamics & Energy Partition

Rolling motion is the fundamental bridge uniting translational and rotational dynamics. When a round object (cylinder, sphere, hoop) rolls without slipping along a surface, the point of contact instantaneously does not move relative to the surface.

5.1 Kinematic Constraints of Pure Rolling

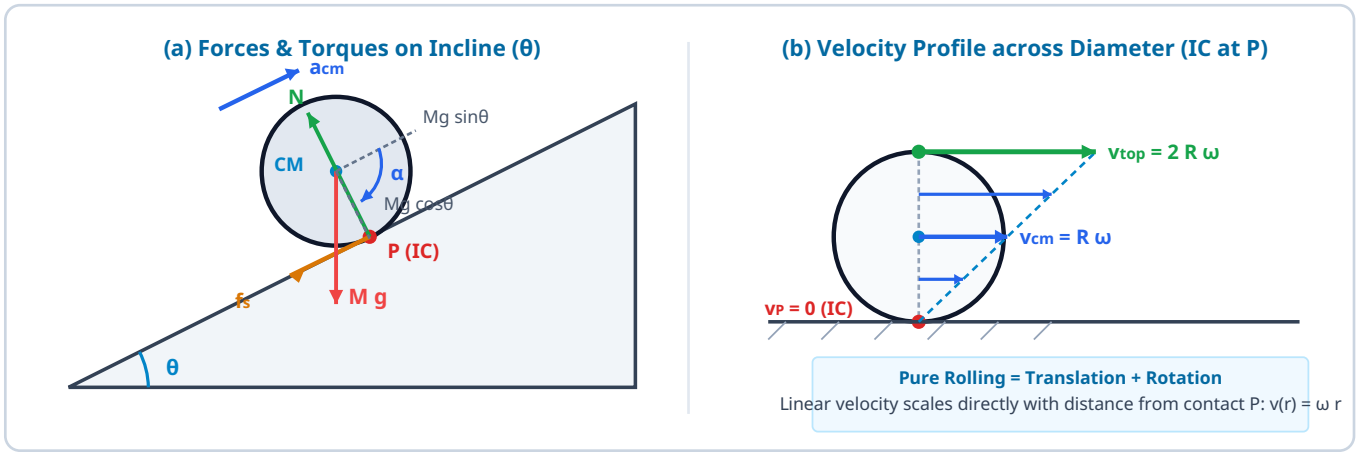
$$\text{Center of Mass Velocity:} \quad v_{cm} = R\,\omega$$

$$\text{Center of Mass Acceleration:} \quad a_{cm} = R\,\alpha$$

Pure rolling can be understood through two completely equivalent physical perspectives:

1. **Superposition of Translation and Rotation:** Pure translational motion of the CM with speed v_{cm} plus pure rotation about the CM with angular speed ω.
2. **Pure Instantaneous Rotation about the Contact Point P:** At any instant, the contact point P has zero instantaneous velocity ($v_P = 0$). The entire body is in pure rotation about an axis through P (the **Instantaneous Center of Zero Velocity**, IC).

Figure 2: Dynamics of Rolling Without Slipping on an Incline & Instantaneous Velocity Field



5.2 Total Kinetic Energy & Incline Acceleration

By König's theorem, the total kinetic energy of a rolling body is the sum of translational and rotational components:

$$K_{total} = K_{trans} + K_{rot} = \tfrac{1}{2} M v_{cm}^2 + \tfrac{1}{2} I_{cm} \omega^2$$

Substituting the pure rolling constraint $\omega = v_{cm} / R$ and defining the dimensionless geometric factor $c = I_{cm} / (M R^2)$:

$$K_{total} = \tfrac{1}{2} M v_{cm}^2 \left[1 + \frac{I_{cm}}{M R^2} \right] = \tfrac{1}{2} M v_{cm}^2 (1 + c)$$

Applying conservation of mechanical energy for a body rolling down a vertical drop h:

$$M g h = \tfrac{1}{2} M v_{cm}^2 (1 + c) \quad \Rightarrow \quad v_{cm} = \sqrt{\left[\frac{2 g h}{1 + c} \right]}$$

Differentiating with respect to time along an incline of angle θ (where $dh/dt = v_{cm} \sin\theta$ and $dv_{cm}/dt = a_{cm}$):

$$a_{cm} = \frac{g \sin\theta}{1 + c} = \frac{g \sin\theta}{1 + \frac{I_{cm}}{M R^2}}$$

5.3 Static Friction & Condition to Prevent Slipping

Newton's equations for rolling down an incline: Parallel to incline: $M g \sin\theta - f_s = M a_{cm}$. Torque about CM: $\tau = f_s R = I_{cm} \alpha = I_{cm} (a_{cm} / R)$. Solving for static friction:

$$f_s = M g \sin\theta \left[\frac{c}{1 + c} \right] = M g \sin\theta \left[\frac{I_{cm}}{I_{cm} + M R^2} \right]$$

Because static friction must not exceed its maximum threshold $f_s \le \mu_s N = \mu_s M g \cos\theta$:

$$\mu_s \ge \frac{c}{1 + c} \tan\theta = \frac{\tan\theta}{1 + \frac{M R^2}{I_{cm}}}$$

⚡ FAST-TRACK 3 **The Universal Rolling Race Shortcut & Mass-Radius Invariance** Instant Ranking

Fundamental Physical Truth: Acceleration a_{cm} and final speed v_{cm} are **completely independent** of the object's mass M and radius R! Only the dimensionless shape factor $c = I_{cm} / (M R^2)$ matters!

⚡ **Fast Ranking Rule: Smallest c Wins Every Race!**

Rank 1 (Winner)	Solid Sphere	c = ⅖ = 0.400	**a = ⁵⁄₇ g sin θ ≈ 0.714 g sin θ**
Rank 2	Solid Disk / Cylinder	c = ½ = 0.500	**a = ⅔ g sin θ ≈ 0.667 g sin θ**
Rank 3	Spherical Shell	c = ⅔ ≈ 0.667	**a = ⅗ g sin θ = 0.600 g sin θ**
Rank 4 (Last)	Hoop / Hollow Ring	c = 1.000	**a = ½ g sin θ = 0.500 g sin θ**

A marble of mass 5 grams will beat a 500-ton hollow steel pipe down any incline, every single time!

Chapter 6: Angular Momentum & The Conservation Principle

Linear momentum **p** = m**v** is the conserved quantity under translational spatial invariance (Noether's Theorem). Under rotational spatial invariance, the corresponding conserved quantity is **Angular Momentum L**.

6.1 Angular Momentum of a Particle & A Rigid Body

For a point particle of linear momentum **p** = m**v** at position **r** relative to origin O, and for an extended rigid body rotating symmetrically about a principal axis:

Particle: $\mathbf{L} = \mathbf{r} \times \mathbf{p} = m(\mathbf{r} \times \mathbf{v})$ [J · s] | **Extended Rigid Body: $L = I\omega$**

6.2 Fundamental Dynamic Equation: $\sum\tau = dL / dt$

Taking the time derivative of **L**:

$\frac{d\mathbf{L}}{dt} = \frac{d\mathbf{r}}{dt} \times \mathbf{p} + \mathbf{r} \times \frac{d\mathbf{p}}{dt} = (\mathbf{v} \times m\mathbf{v}) + (\mathbf{r} \times \mathbf{F}_{net})$. Since $\mathbf{v} \times \mathbf{v} = 0$ identically and $\mathbf{r} \times \mathbf{F}_{net} = \boldsymbol{\tau}_{net}$:

$$\sum \boldsymbol{\tau}_{ext} = \frac{d\mathbf{L}}{dt}$$

6.3 The Law of Conservation of Angular Momentum

▶ Conservation Principle

If the net external torque acting on a system about a given origin is zero ($\sum \boldsymbol{\tau}_{ext} = \mathbf{0}$), the total angular momentum remains strictly constant in time:

$$L_{total} = \text{const} \quad \Rightarrow \quad L_{initial} = L_{final} \quad \Rightarrow \quad I_i \omega_i = I_f \omega_f$$

⚡ FAST-TRACK 4 **Angular Momentum Scaling & Kinetic Energy Amplification** Ratio Rule

Crucial Physics Paradox: When angular momentum is conserved, is rotational kinetic energy conserved? **NO!** Kinetic energy increases when I decreases!

⚡ **Energy Scaling Formula:** Express K_{rot} in terms of conserved momentum L:

$$K_{rot} = \tfrac{1}{2} I \omega^2 = \frac{L^2}{2I} \quad \Rightarrow \quad \frac{K_f}{K_i} = \frac{I_i}{I_f} = \frac{\omega_f}{\omega_i}$$

Figure Skater Example: If a skater pulls in her arms reducing I by factor of 3 ($I_f = \frac{1}{3} I_i$):

- Spin speed triples: $\omega_f = 3\,\omega_i$.
- Kinetic energy triples: $K_f = 3\,K_i$.

Where does the extra kinetic energy $\Delta K = 2K_i$ come from? From **internal muscular work** performed by the skater pulling her arms inward against the centrifugal inertia!

6.4 Astrophysical Applications: Stellar Core Collapse & Pulsar Formation

The most dramatic manifestation of angular momentum conservation in nature occurs during the death of massive stars. When a star exhausts its nuclear fuel, its central core undergoes catastrophic gravitational collapse. Consider an iron stellar core of initial radius $R_i \approx 5.0 \times 10^6$ m rotating once every 30 days ($\omega_i = 2\pi / (30 \times 86400\ \text{s}) \approx 2.4 \times 10^{-6}$ rad/s). In less than a second, gravity collapses this core into a dense neutron star of radius $R_f \approx 12.0$ km $= 1.2 \times 10^4$ m.

$$\mathbf{I_i\,\omega_i = I_f\,\omega_f \quad \Rightarrow \quad \omega_f = \omega_i \left[\frac{R_i}{R_f}\right]^2 = (2.4 \times 10^{-6}\ \text{rad/s}) \left[\frac{5.0 \times 10^6\ \text{m}}{1.2 \times 10^4\ \text{m}}\right]^2 \approx 416\ \text{rad/s} \approx 66.2\ \text{rev/s!}}$$

The collapsed stellar core becomes a rapidly rotating **neutron star (pulsar)** emitting focused beams of synchrotron radiation 66 times each second! Furthermore, by Fast-Track 4, the core's rotational kinetic energy is amplified by:

$$\frac{K_f}{K_i} = \frac{I_i}{I_f} = \left[\frac{R_i}{R_f}\right]^2 \approx 1.7 \times 10^5$$

This immense surge in kinetic energy ($\sim 10^{44}$ Joules) is supplied directly by the catastrophic release of gravitational potential energy during collapse.

6.5 Internal Couplings: Helicopter Anti-Torque & The Falling Cat Reflex

The Helicopter Anti-Torque Problem: When a helicopter engine applies an internal torque to spin its main rotor blades clockwise, Newton's third law dictates an equal and opposite counter-torque acting on the helicopter fuselage. Without external torque, the fuselage would spin violently counter-clockwise to conserve total system angular momentum ($\mathbf{L} = \mathbf{0}$). To maintain yaw equilibrium, an auxiliary tail rotor produces a sideways thrust force F_{tail} at tail boom distance d, supplying an external aerodynamic torque $\tau_{ext} = F_{tail}\,d$ that exactly cancels the main rotor's reactionary torque.

The Cat Righting Reflex (Zero-Angular-Momentum Reorientation): A cat dropped upside-down from rest with zero initial angular momentum ($\mathbf{L} = \mathbf{0}$) consistently rights itself to land safely on its feet. It accomplishes this remarkable feat with zero external torque ($\sum\tau_{ext} = 0$) by exploiting non-holonomic internal body mechanics:

1. The cat arches and bends its spine, effectively dividing its body into two independent segments: a front half and a rear half.
2. It tucks its front legs tightly (greatly minimizing I_{front}) while extending its rear legs outward (greatly maximizing I_{rear}).
3. It twists its front and rear segments in opposite directions. By conservation of angular momentum:

   $$\mathbf{I_{front}\,\omega_{front} + I_{rear}\,\omega_{rear} = 0 \quad \Rightarrow \quad \omega_{front} = -\left[\frac{I_{rear}}{I_{front}}\right]\omega_{rear}}$$

   Because $I_{rear} \gg I_{front}$, the front torso rotates through a large positive angle (+90°) while the rear only counter-rotates through a tiny negative angle (−15°).
4. The cat then reverses its limbs: extending front legs and tucking rear legs, then twisting back. The net result is a complete 180° righting turn while total $\mathbf{L} = \mathbf{0}$ remains strictly conserved at every microsecond!

Part IV: 3D Vector Dynamics & Gyroscopes

Chapter 7: 3D Vector Dynamics & Gyroscopic Precession

In general 3D space, angular velocity **ω** and angular momentum **L** are not necessarily parallel. They are linked by the 3 × 3 **Inertia Tensor I**:

$$\mathbf{L}_i = \sum_j \mathbf{I}_{ij}\,\boldsymbol{\omega}_j, \quad \text{where} \quad I_{xx} = \int (y^2 + z^2)\,dm, \quad I_{xy} = -\int x\,y\,dm$$

Only when a body rotates about one of its three orthogonal **Principal Axes of Inertia** (where the off-diagonal products of inertia vanish) does **L** point in the exact same direction as **ω**.

7.1 Mechanics of Gyroscopic Precession

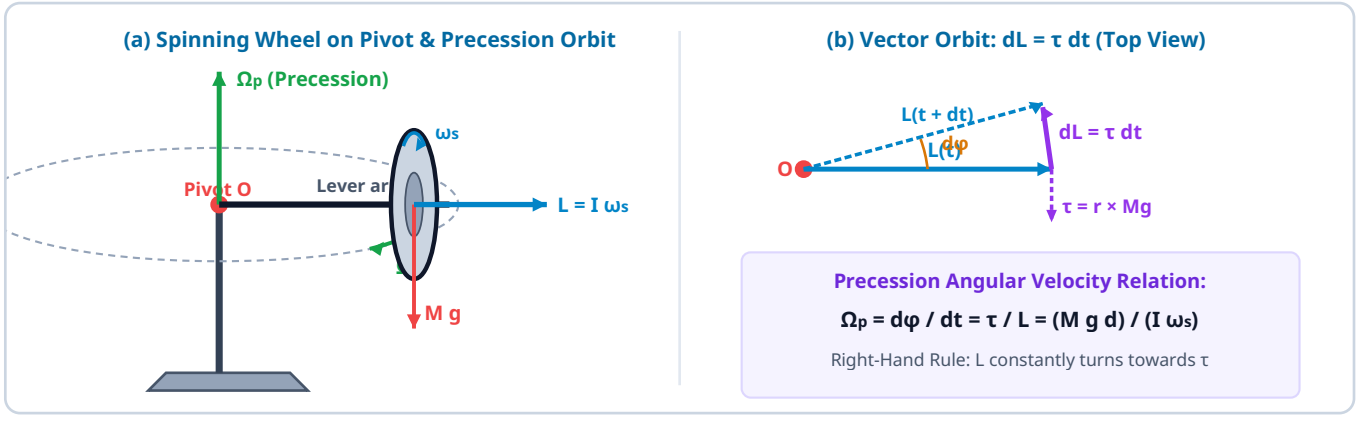
Consider a symmetric rotor of moment of inertia I spinning rapidly at rate ω_s (angular momentum $\mathbf{L} = I\,\boldsymbol{\omega}_s$), supported horizontally at a pivot O at center-of-mass distance d. Gravity exerts downward force M**g**, producing horizontal torque:

$$\boldsymbol{\tau} = \mathbf{r} \times M\mathbf{g} \quad \Rightarrow \quad \tau = M\,g\,d, \quad \text{with} \quad \boldsymbol{\tau} \perp \mathbf{L}$$

By Newton's rotational law $\boldsymbol{\tau} = d\mathbf{L}/dt$, the change $d\mathbf{L} = \boldsymbol{\tau}\,dt$ is strictly perpendicular to **L**. Torque cannot alter the spin magnitude $|\mathbf{L}|$; it continually redirects **L** horizontally, sweeping an angle dφ in time dt:

$$|d\mathbf{L}| = L\,d\varphi = \tau\,dt = (M\,g\,d)\,dt \quad \Rightarrow \quad \Omega_p = \frac{d\varphi}{dt} = \frac{\tau}{L} = \frac{M\,g\,d}{I\,\omega_s}$$

Figure 3: Gyroscopic Precession Mechanics & Vector Angular Momentum Torque Coupling



7.2 Vector Precession Relation & Stability

In full vector notation, the torque relates to precession as:

$$\boldsymbol{\tau} = \boldsymbol{\Omega}_p \times \mathbf{L}$$

Notice the remarkable inverse dependence: the faster the rotor spins ($\omega_s \uparrow$), the *slower* it precesses ($\Omega_p \downarrow$)! If the spinning wheel is tilted at an angle θ relative to the vertical, the torque becomes $\tau = M\,g\,d\sin\theta$ while the horizontal component of angular momentum is $L\sin\theta$. The $\sin\theta$ terms cancel identically, leaving $\Omega_p = (M\,g\,d)\,/\,(I\,\omega_s)$ independent of the inclination angle θ!

7.3 Euler's Equations of Motion & The Dzhanibekov Effect

In a rotating body-fixed coordinate frame aligned with the principal axes of inertia (where principal moments of inertia satisfy $I_1 \le I_2 \le I_3$), Euler's dynamical equations for torque-free rotation ($\sum \boldsymbol{\tau} = \mathbf{0}$) take the celebrated form:

$$I_1 \frac{d\omega_1}{dt} - (I_2 - I_3)\,\omega_2\,\omega_3 = 0, \quad I_2 \frac{d\omega_2}{dt} - (I_3 - I_1)\,\omega_3\,\omega_1 = 0, \quad I_3 \frac{d\omega_3}{dt} - (I_1 - I_2)\,\omega_1\,\omega_2 = 0$$

The Intermediate Axis Theorem (Tennis Racket Instability): Linear stability perturbation analysis yields a profound physical revelation:

- **Major Axis (I_3) & Minor Axis (I_1):** Perturbations induce bounded, purely sinusoidal oscillations. Rotation about either extreme axis is strictly **Lyapunov stable**.
- **Intermediate Axis (I_2):** The product of coefficients $(I_3 - I_1)(I_1 - I_2)$ is strictly positive, driving exponential perturbation growth: $d^2(\delta\omega_1)/dt^2 > 0$. Rotation about the intermediate principal axis is **unconditionally unstable**!

In microgravity, any rigid body spun about its intermediate axis (such as a wing nut, T-handle, or flipped tennis racket) spontaneously flips by 180° back and forth in a periodic cyclic inversion. This striking phenomenon—the **Dzhanibekov effect**—is an exact topological requirement imposed by the intersection of the constant-energy ellipsoid ($\mathbf{2E = \sum I_i\,\omega_i^2}$) and the constant angular momentum sphere ($\mathbf{L^2 = \sum I_i^2\,\omega_i^2}$).

⚡ FAST-TRACK 5 **The 3-Second Gyroscopic Precession Shortcut & Chase Rule** Instant Vector Rule

Exam Confusion: Students frequently struggle to remember which way a gyroscope precesses or get entangled in sign conventions.

⚡ **The Universal Chase Rule: L Constantly Chases τ!**

1. Point your right thumb along the spin vector **L**.
2. Calculate the torque vector $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ using your right hand.
3. The tip of **L** must always pivot towards **τ**! Because τ moves as L rotates, L chases τ in an endless circular loop.
4. Precession frequency is always: $\boldsymbol{\Omega_p}$ **= Torque / Angular Momentum =** $\boldsymbol{\tau / L}$ in 3 seconds!

Part V: Step-by-Step Worked Problems & Reference Appendices

Chapter 8: 4 Complete Step-by-Step Worked Problems

Problem 1: Rotational Kinematics & Flywheel Braking Dynamics **Chapter 1, 3 & Fast-Track 1**

Problem Statement: A solid disk flywheel of mass M = 40.0 kg and radius R = 0.500 m is spinning freely at ω_0 = 1200 rpm. A constant frictional braking torque of τ_{brake} = 25.0 N·m is engaged to bring the flywheel to rest.

Given Parameters: M = 40.0 kg, R = 0.500 m, ω_0 = 1200 rpm = 1200 × (2π/60) = 40π rad/s ≈ 125.66 rad/s, τ_{brake} = 25.0 N·m.

Step 1: Moment of Inertia & Angular Acceleration:

For a uniform solid disk, $I = \frac{1}{2} M R^2 = \frac{1}{2}(40.0\ kg)(0.500\ m)^2$ = **5.00 kg · m²**.

By Newton's second law for rotation, $\tau = I\alpha \Rightarrow \alpha = -\tau_{brake} / I = -(25.0\ N\cdot m) / (5.00\ kg\cdot m^2)$ = **−5.00 rad/s²**.

Step 2: Stopping Time:

Using $\omega(t) = \omega_0 + \alpha t = 0 \Rightarrow t_{stop} = \omega_0 / |\alpha|$ = (40π rad/s) / (5.00 rad/s²) = **8π s ≈ 25.13 s**.

Step 3: Total Revolutions to Rest:

Total angular displacement: $\Delta\theta = \omega_0^2 / (2|\alpha|) = (40\pi)^2 / (2 \times 5.00) = 1600\pi^2 / 10.0 = 160\pi^2$ rad ≈ 1579.14 rad.

Total revolutions: $N_{rev} = \Delta\theta / (2\pi) = 160\pi^2 / (2\pi)$ = **80π rev ≈ 251.33 revolutions**.

Step 4: Accelerations on Rim at t = 10.0 s:

ω(10) = 125.66 − 5.00(10.0) = 75.66 rad/s.

Tangential acceleration: $a_t = |\alpha| R = (5.00\ rad/s^2)(0.500\ m)$ = **2.50 m/s²**.

Centripetal acceleration: $a_c = \omega^2 R = (75.66\ rad/s)^2(0.500\ m)$ = **2862.5 m/s²**.

Total linear acceleration: $a_{total} = \sqrt{(a_t^2 + a_c^2)} = \sqrt{(2.50^2 + 2862.5^2)} \approx$ **2862.5 m/s²** (completely dominated by centripetal pull).

Sanity / Work-Energy Check:

Work done by braking torque: $W = -\tau_{brake} \Delta\theta = -(25.0\ N\cdot m)(1579.14\ rad) = -39478.5$ J.

Initial kinetic energy: $K_0 = \frac{1}{2} I \omega_0^2 = \frac{1}{2}(5.00\ kg\cdot m^2)(125.664\ rad/s)^2 = 39478.5$ J.

Since $\Delta K = K_{final} - K_0 = 0 - 39478.5\ J = W_{brake}$, energy conservation holds with 100% precision!

Problem 2: Coupled Atwood System with Massive Disk Pulley — Chapter 3 & Fast-Track 2

Problem Statement: Two masses m_1 = 4.00 kg and m_2 = 2.00 kg are connected by a light string over a frictionless axle supporting a solid disk pulley of mass M_p = 3.00 kg and radius R = 0.200 m. The string does not slip. Determine the linear acceleration a, angular acceleration α, and string tensions T_1 and T_2 on either side.

Given Parameters: m_1 = 4.00 kg, m_2 = 2.00 kg, M_p = 3.00 kg, R = 0.200 m, g = 9.80 m/s^2.

Step 1: Application of Fast-Track 2 (Equivalent Mass):
The solid disk pulley has rotational inertia $I = \frac{1}{2} M_p R^2$. Its effective translational inertia is: $m_{eff} = I / R^2 = \frac{1}{2} M_p$ = ½ (3.00 kg) = **1.50 kg**.
The net driving gravitational force is $(m_1 - m_2)\, g$.

Step 2: Linear & Angular Acceleration:
Using the Fast-Track 2 master equation:

$$\mathbf{a = \frac{(m_1 - m_2)\, g}{m_1 + m_2 + \frac{1}{2} M_p} = \frac{(4.00 - 2.00)(9.80)}{4.00 + 2.00 + 1.50} = \frac{19.60}{7.50} = 2.6133\ m/s^2 \approx 2.61\ m/s^2}$$

Angular acceleration of the pulley: $\alpha = a / R = (2.6133\ m/s^2) / (0.200\ m)$ = **13.07 rad/s^2**.

Step 3: Calculating Distinct String Tensions:
For m_1 (accelerating downwards): $m_1 g - T_1 = m_1 a \Rightarrow T_1 = m_1(g - a) = 4.00(9.80 - 2.6133)$ = **28.75 N**.
For m_2 (accelerating upwards): $T_2 - m_2 g = m_2 a \Rightarrow T_2 = m_2(g + a) = 2.00(9.80 + 2.6133)$ = **24.83 N**.

Step 4: Torque Consistency Verification:
Net torque on pulley: $\tau_{net} = (T_1 - T_2) R = (28.75 - 24.83)(0.200) = (3.92\ N)(0.200\ m) = 0.784\ N \cdot m$.
$I \alpha = [\frac{1}{2}(3.00)(0.200)^2](13.07) = (0.0600\ kg \cdot m^2)(13.07\ rad/s^2) = 0.784\ N \cdot m$. Exact agreement!

Physical Sanity Check:
In the limit of a massless pulley ($M_p \to 0$), $a_{ideal} = (2/6)(9.80) = 3.27\ m/s^2$. Because the real pulley has inertia (M_p = 3.0 kg), it resists angular acceleration, yielding $a = 2.61\ m/s^2 < 3.27\ m/s^2$, exactly as physics dictates.

Problem 3: The Incline Rolling Race: Solid Cylinder vs. Spherical Shell — Chapter 5 & Fast-Track 3

Problem Statement: A solid cylinder (Body A: M_A = 2.00 kg, R_A = 0.150 m) and a thin spherical shell (Body B: M_B = 1.20 kg, R_B = 0.100 m) start from rest and roll without slipping down a 30.0° incline of length L = 5.00 m (vertical height h = L sin 30° = 2.50 m). Determine: (a) linear acceleration of each, (b) final speed at bottom, (c) time to finish and arrival gap, and (d) minimum static friction μ_s required.

Given Parameters: θ = 30.0°, L = 5.00 m, h = 2.50 m, g = 9.80 m/s^2. Solid cylinder c_A = 0.500; Spherical shell c_B = ⅔ ≈ 0.667.

Step 1: Accelerations (Fast-Track 3):

For Solid Cylinder A: $a_A = (g \sin\theta) / (1 + c_A) = (9.80 \times 0.500) / (1 + 0.500) = 4.90 / 1.50 =$ **3.267 m/s^2**.

For Spherical Shell B: $a_B = (g \sin\theta) / (1 + c_B) = (9.80 \times 0.500) / (1 + 0.667) = 4.90 / (5/3) =$ **2.940 m/s^2**.

Step 2: Speed at the Bottom:

For Cylinder A: $v_A = \sqrt{(2\, a_A\, L)} = \sqrt{[2(3.2667)(5.00)]} = \sqrt{32.667} =$ **5.715 m/s**.

For Spherical Shell B: $v_B = \sqrt{(2\, a_B\, L)} = \sqrt{[2(2.940)(5.00)]} = \sqrt{29.40} =$ **5.422 m/s**.

Step 3: Finishing Times & Race Gap:

$t_A = \sqrt{(2\, L / a_A)} = \sqrt{[2(5.00) / 3.2667]} = \sqrt{3.0612} =$ **1.750 s**.

$t_B = \sqrt{(2\, L / a_B)} = \sqrt{[2(5.00) / 2.940]} = \sqrt{3.4014} =$ **1.844 s**.

Time difference: $\Delta t = t_B - t_A = 1.844 - 1.750 =$ **0.094 s ≈ 94 ms**. The solid cylinder convincingly wins!

Step 4: Minimum Static Friction Thresholds:

Formula: $\mu_{s,min} = [c / (1 + c)] \tan\theta = [c / (1 + c)] \tan 30° = [c / (1 + c)] \times 0.5774$.

For Cylinder A: $\mu_{s,min,A} = (0.500 / 1.500)(0.5774) = ⅓ \times 0.5774 =$ **0.192**.

For Spherical Shell B: $\mu_{s,min,B} = [(⅔) / (⅗)](0.5774) = ⅖ \times 0.5774 =$ **0.231**.

If the surface has μ_s = 0.25, both objects roll smoothly without slipping ($\mu_s > \mu_{s,min}$).

Mass-Radius Invariance Sanity Check: Notice that the masses (2.0 kg vs 1.2 kg) and radii (0.15 m vs 0.10 m) never entered any kinematic calculation. A tiny 10-gram cylinder would beat a 1000-kg hollow sphere by the exact same 94 ms margin!

Problem 4: Ballistic Bullet Impact on Pivoted Pendulum Rod — Chapter 6 & Fast-Track 4

Problem Statement: A uniform thin rod of mass M = 1.50 kg and length L = 0.800 m is suspended vertically from a frictionless pivot at its top end. A high-speed bullet of mass m = 25.0 g = 0.0250 kg moving horizontally at v_0 strikes the bottom tip of the rod (x = L) and embeds itself instantly. Determine: (a) system rotational inertia, (b) initial angular velocity ω_0 after impact for v_0 = 90.0 m/s, (c) mechanical energy lost, and (d) maximum swing angle θ_{max}, plus threshold speed to complete full vertical loops.

Given Parameters: M = 1.50 kg, L = 0.800 m, m = 0.0250 kg, g = 9.80 m/s^2.

Step 1: Total Moment of Inertia about Pivot:

$I_{rod,end} = \frac{1}{3} M L^2 = \frac{1}{3} (1.50 \text{ kg})(0.800 \text{ m})^2 = 0.50 \times 0.640 = 0.320 \text{ kg·m}^2$.

$I_{bullet} = m L^2 = (0.0250 \text{ kg})(0.800 \text{ m})^2 = 0.0160 \text{ kg·m}^2$.

$I_{total} = I_{rod,end} + I_{bullet} = 0.320 + 0.0160 = \mathbf{0.3360 \text{ kg} \cdot \text{m}^2}$.

Step 2: Angular Momentum Conservation about Pivot:
During the brief collision, the pivot exerts a strong impulse force, so linear momentum is NOT conserved. However, the pivot force exerts zero torque about the pivot axis. Thus, angular momentum about the pivot is strictly conserved:

$$L_{before} = m v_0 L = (0.0250 \text{ kg})(90.0 \text{ m/s})(0.800 \text{ m}) = \mathbf{1.800 \text{ kg} \cdot \text{m}^2 / \text{s}}$$

$$\omega_0 = \frac{L_{before}}{I_{total}} = \frac{1.800 \text{ kg} \cdot \text{m}^2 / \text{s}}{0.3360 \text{ kg} \cdot \text{m}^2} = \mathbf{5.357 \text{ rad/s}}$$

Step 3: Energy Loss in Inelastic Collision:

$K_{initial} = \frac{1}{2} m v_0^2 = \frac{1}{2} (0.0250 \text{ kg})(90.0 \text{ m/s})^2 = \mathbf{101.25 \text{ J}}$.

$K_{after} = \frac{1}{2} I_{total} \omega_0^2 = \frac{1}{2} (0.3360 \text{ kg·m}^2)(5.357 \text{ rad/s})^2 = \mathbf{4.821 \text{ J}}$.

Energy lost to deformation/heat: $\Delta E = 101.25 - 4.82 = \mathbf{96.43 \text{ J (95.2\% dissipated)}}$.

Step 4: Maximum Swing Angle θ_{max}:

Center of mass distance y_{cm} from pivot: $y_{cm} = [M(L/2) + m L] / (M + m) = [1.50(0.400) + 0.0250(0.800)] / 1.525 = 0.620 / 1.525 = 0.4066$ m.

By mechanical energy conservation during upswing: $K_{after} = (M + m) g y_{cm} (1 - \cos \theta_{max})$.

$4.821 \text{ J} = (1.525 \text{ kg})(9.80 \text{ m/s}^2)(0.4066 \text{ m})(1 - \cos \theta_{max}) = 6.076 \text{ J} \times (1 - \cos \theta_{max})$.

$1 - \cos \theta_{max} = 4.821 / 6.076 = 0.7934 \Rightarrow \cos \theta_{max} = 0.2066 \Rightarrow \mathbf{\theta_{max} = \arccos(0.2066) = 78.07°}$.

Looping Threshold Insight: To complete a full 360° over-the-top loop, the pendulum requires $K_{after} \geq 2 (M+m) g y_{cm} = 2(6.076) = 12.15$ J. Setting $\frac{1}{2} I_{total} \omega_{loop}^2 = 12.15$ J yields $\omega_{loop} = 8.51$ rad/s, corresponding to a bullet speed of $v_{loop} \geq 143$ m/s. For military rifle bullets ($v_0 \approx 700–900$ m/s), the rod executes dozens of rapid over-the-top loops!

Chapter 9: Master Rotational Dynamics Cheat Sheet & Exam Checklist

9.1 The Grand Translation vs. Rotation Equivalence Matrix

Translational Physics Concept	Linear Equation	Rotational Counterpart	Rotational Equation
Linear Displacement	x [m]	Angular Displacement	$\theta = s / r$ [rad]
Linear Velocity	$v = dx / dt$ [m/s]	Angular Velocity	$\omega = d\theta / dt$ [rad/s]
Linear Acceleration	$a = dv / dt$ [m/s^2]	Angular Acceleration	$\alpha = d\omega / dt$ [rad/s^2]
Inertia (Mass)	m [kg]	Rotational Inertia	$I = \int r_{\perp}^2 \, dm$ [kg · m^2]
Force	$\mathbf{F} = m\,\mathbf{a}$ [N]	Torque	$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = I\,\boldsymbol{\alpha}$ [N · m]
Kinetic Energy	$K = \frac{1}{2} m v^2$ [J]	Rotational Kinetic Energy	$K_{rot} = \frac{1}{2} I \omega^2$ [J]
Work	$W = \int F \, dx$ [J]	Rotational Work	$W = \int \tau \, d\theta$ [J]
Instantaneous Power	$P = \mathbf{F} \cdot \mathbf{v}$ [W]	Rotational Power	$P = \boldsymbol{\tau} \cdot \boldsymbol{\omega}$ [W]
Linear Momentum	$\mathbf{p} = m\,\mathbf{v}$	Angular Momentum	$\mathbf{L} = \mathbf{r} \times \mathbf{p} = I\,\boldsymbol{\omega}$
Newton's 2nd Law (General)	$\sum \mathbf{F} = d\mathbf{p} / dt$	Rotational 2nd Law (General)	$\sum \boldsymbol{\tau} = d\mathbf{L} / dt$
Momentum Conservation	If $\sum \mathbf{F}_{ext} = 0 \Rightarrow \Delta\mathbf{p} = 0$	Angular Conservation	If $\sum \boldsymbol{\tau}_{ext} = 0 \Rightarrow \Delta\mathbf{L} = 0$

9.2 Essential Examination Pitfalls & Sanity Checklist

▶ 5 Rules to Never Lose Points on Rotational Dynamics

1. **Radian Measure is Mandatory:** In all kinematic formulas connecting linear and rotational variables ($v = \omega r$, $a = \alpha r$, $\theta = s/r$), ω and α MUST be expressed in radians per second (rad/s), NEVER degrees or rpm! Multiply rpm by $2\pi/60$.
2. **Momentum Pivot Consistency:** When computing angular momentum or torque $\sum \boldsymbol{\tau} = d\mathbf{L}/dt$, ALL torques and angular momenta MUST be evaluated with respect to the EXACT same reference origin O! Mixing pivot origins in a single equation invalidates the solution.
3. **Static Friction Does ZERO Work in Pure Rolling:** In rolling without slipping, the contact point is instantaneously at rest ($v_{contact} = 0$). Therefore, static friction dissipates zero mechanical energy: $dW = f_s \cdot ds = 0$. Mechanical energy is 100% conserved!
4. **Torque Lever Arm Orthogonality:** Torque is $\tau = r F \sin\varphi$. Always identify the perpendicular distance $r_{\perp}$ to the line of force. A radial force passing directly through the pivot produces zero torque (lever arm = 0).
5. **Rigid Body Conservation Condition:** $\Delta\mathbf{L} = 0$ requires $\sum \tau_{ext} = 0$. If a pivot exerts a support force, linear momentum is destroyed, but angular momentum about that pivot is strictly conserved!